

Research Paper

Experimental study on the thermal management performance of lithium-ion battery with PCM combined with 3-D finned tube

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ABSTRACT

As an important intermediary between the green energy and human society, the lithium-ion battery has promising prospects in the new energy vehicles, energy storage, and green development fields. However, lithium-ion batteries can generate a large amount of heat during operation. In addition, excess temperature or big temperature difference of the surface of the lithium-ion battery can reduce its performance, which also results in reducing its operational life and even triggers incidents such as fires and explosions. Therefore, studying efficient thermal management methods for lithium-ion batteries is crucial for increasing their application range. This paper proposes a novel thermal management method by combining 3-D finned tubes with PCM. The thermal management system maintains a battery operating temperature less than 45 °C and a maximum temperature difference less than 2.4 °C. The influence of different fin structure parameters on the melting dynamics of PCM and the heat transfer performance is then discussed. The obtained results show that under each solid-liquid ratio of PCM, the heat transfer performance increases with the increases of the fin height, while it initially decreases and then increases with the increase of the width and axial spacing. Moreover, the heat transfer performance of the 3-D finned tube with different PCM liquid-solid ratios is investigated. The obtained results demonstrate that for constant fin parameters, the heat transfer reaches its highest performance for a PCM liquid-solid ratio greater than 2:2. The findings of this study lay a foundation for the design of efficient thermal management systems incorporating PCM applications.

1. Introduction

As countries around the world prioritize the development of green energy sources and address concerns about fossil fuel depletion and climate change, the lithium-ion battery have become a widely used source of power for various applications, including electric vehicles, energy storage systems, communication equipment, and engineering machinery [1]. However, due to the existence of Joule heating, reversible heat, and irreversible heat, the high temperature and poor low temperature uniformity problems generated by the lithium-ion batteries under different charging and discharging power shorten their service life, and even cause fire and explosion [2,3]. Wang et al. [4] demonstrated that a temperature greater than 50 °C can significantly reduce the cycle life and charging efficiency of lithium-ion battery. It can also

lead to inconsistencies in the capacity of the battery pack, and even cause electrical accidents. Therefore, the development of advanced lithium-ion battery thermal management systems (BTMs) is essential for improving the safety of usage and storage [5].

Typically, the effective BTMS should reduce the maximum temperature while minimizing the temperature disparity [6,7]. The temperature control technology initiated on phase change materials (PCM) outperforms conventional methodologies, such as active air cooling and circulating liquid cooling [7], in terms of temperature regulation. Moreover, it represents the attributes of environmental sustainability and efficient heat exchange [8,9]. Recent studies highlighted the growing prominence of the advantages brought by the temperature control technology based on PCM. They demonstrated that this method has high heat exchange efficiency, high temperature uniformity, and low cost [10–12]. Xie et al. [13] deduced that the use of PCM-controlled

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Nomenclature		q heat flux ($\text{W}/\text{m}^2(- -)$)	
A	area of tube surface (m^2)	Acronyms	
C	charge/discharge rate (-)	3-D	Three-dimensional
C_p	heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)	BTMS	Battery Thermal Management System
H	fin height(mm)	h	Hour
K	heat transfer coefficient ($\text{W}/\text{m}^2(- -) \text{K}^{-1}$)	PCM	Phase Change Material
Pa	axial spacing (mm)	min	Minute
P_c	circumferential spacing (mm)	Subscript	
R_x	thermal resistance ($\text{m}^2 \text{K W}^{-1}$)	<i>cond</i>	thermal conduction
T_x	temperature (K)	<i>conv</i>	thermal convection
W	fin width(mm)	<i>face</i>	solid-liquid interface of PCM
Greek symbols		<i>tube</i>	tube surface
ρ	density (kg m^{-3})	<i>wall</i>	heated plate wall
λ	thermal conductivity($\text{W m}^{-1} \text{K}^{-1}$)		

battery modules results in more uniform temperature at different discharge rates compared with air-cooled batteries, which proves the good effect of the PCM temperature control. Javani et al. [14] conducted a study on the cooling efficacy of PCM in batteries generating heat at varying volumetric rates. Their results showed that PCM is able to mitigate the battery peak temperature, which leads to improved temperature homogeneity.

However, the PCM has low thermal conductivity, and the large amount of internally stored heat cannot be quickly and effectively transferred to the external environment [15]. The PCM-based cooling technology may fail, especially in long-term operation, when dealing with high power and high current, and in some extreme temperature environments [16]. Therefore, many researchers focus on the improvement of the heat transfer capacity of the PCM and the quick dissipation of its heat, which aims at ensuring its long-term stable operation [17]. For instance, Singh et al. [18] developed a dual BTMS that tackles the cooling ability of PCM in conjunction with forced air for lithium-ion battery. This was compared with a PCM-based BTMS incorporating a composite of paraffin and expanded graphite (PA/EG). The obtained results showed an enhanced reliability of the hybrid BTMS without increasing the complexity of the system, while effectively avoiding the excessive heat accumulation and maintaining the temperature of the cycling lithium-ion battery below 50 °C. Lv et al. [19] fabricated a tri-component composite material incorporating expanded graphite, paraffin, and low-density polyethylene. This composite was then merged with low fins and connected to the battery module, resulting in advanced heat dissipation efficiency. It effectively dispersed the substantial heat build-up from the PCM into the external air environment. However, the approach of improving the heat transfer efficiency through PCM modification is not suitable for prolonged usage, as the additives usually precipitate (PA/EG) [16]. Some researchers also adopted active cooling strategies, that integrate water cooling to dissipate the heat stored in the PCM [17,20]. Luo et al. [16] compared different cooling schemes for lithium-ion batteries. They deduced that the combination of the strong cooling effect of liquids with the low temperature inconsistency of PCM is the most efficient. On one side of the equation, the liquid cooling quickly extracts the thermal energy accumulated in the PCM. In addition, the PCM maintains a minute fluctuation in the melting point, and it has a high thermal storage capacity, which ensures that the average temperature of the battery remains within the requisite parameters. Heat exchangers with finned structure are widely used in BTMS to enhance the thermal management ability of the combined liquid cooling and PCM [17,20,21]. Liu et al. [17] studied a shell-and-tube battery thermal management system based on liquid cooling. They deduced that the PCM/liquid cooling scheme has high temperature control ability. Moaveni et al. [20] deduced that using

a liquid cooling tube equipped with fins, combined with PCM, can decrease the temperature of the battery by 10.36 °C at a 4C discharge rate. However, due to the limitations of the traditional finned tube manufacturing processes, the contact points between fins produced by welding or extrusion and the base tube have high thermal resistance, which significantly decreases the heat exchange efficiency between the PCM and the heat exchanger [22].

The 3-D finned tube is a type of heat exchangers having high heat transfer performance and widely used in industry [22–24]. In contrast to other types of fin heat exchangers, the 3-D finned tubes feature rotated, discrete, and systematically arranged scaly fins. These fins can be cut into the surface of the base tubes made from various materials using specialized machine tools and cutter heads [25]. Due to their structure and high specific surface area, the 3-D finned tubes can enhance the natural convection and improve the heat transfer efficiency [26,27]. The processing materials have very few limitations, and the fin arrangement and structural parameters can be easily modified. Furthermore, it has no contact thermal resistance at the fin-tube connection [28].

The most significant feature of the 3-D finned tube is that the structural parameters of the fins are closely related to its heat transfer efficiency. Studying the influence of the 3-D finned tube parameters on the heat transfer efficiency can conserve materials and maximize the latent heat advantage of PCM. In addition, when using PCM for thermal management, its complete melting always indicates impending failure in thermal management [18]. It is only practical to keep the PCM in a state of solid-liquid coexistence. It is known that the phase transition is a very complex process, and the PCM melting state can significantly affect the heat transfer performance of the thermal management device. Thus, it is crucial to study the impact of PCM on the heat transfer efficiency in various melting states.

In summary, this study explores the thermal management of lithium-ion batteries through the integration the 3-D finned tube technology with PCM. An experimental study is conducted to analyze the thermal transfer behaviors in PCM across different melting stages. An analysis is then conducted on the heat transfer ability of 3-D finned tube with different fin parameters under different solid-liquid ratios. The changes of thermal conductivity in the solid phase region and convective thermal resistance in the liquid phase region during the phase transition process are also analyzed. Furthermore, an analysis is conducted on the efficiency of the thermal conductivity when the PCM is paired with a 3-D finned tube, under varying melting states in a stable condition. This study facilitates the optimization of the BTMS design and provides new insights into thermal management systems involving the application of PCM.

2. Experiment

2.1. Experimental system

The main equipment of the experimental system, shown in Fig. 1, includes valve, cooling water tank, peristaltic pump, rotor flowmeter, state control device (charge/discharge controller and voltage regulator), experimental chamber containing the heating plates, PCM, 3-D finned tubes, and measuring components. The inlet and outlet are connected through the pipeline to the cooling water tank, the constant temperature water tank, and the high-precision experimental peristaltic pump. The voltage regulator and power meter is used for the regulation and surveillance of the plate heating power. The constant temperature water tank regulates the cooling water temperature, while the peristaltic pump manages its inlet flow. The data collector is applied to collect the temperature of each measurement point in the experimental process. During the experiment, the peristaltic pump delivers the cooling water from the temperature-controlled water tank into the 3-D finned tubes within the experimental chamber. This allows to dissipate the heat generated by the heating plates simulating the batteries. The cooling water flows out and enters the cooling water tank through a valve, and then returns to the temperature-controlled water tank. The melting state of the PCM is observed through the transparent top cover of the experimental chamber. The main equipment used in the experiment and their specific parameters are shown in Table 1.

Before starting the experiment, the liquid PCM is melted and stirred in a water bath. The melted PCM is then poured into the experimental chamber and allowed to cool for 24 h until reaching a stable state. The heating plates are turned on to heat the PCM until the solid-liquid interface approaches the calibration line on the experimental chamber. During this process, flow rate and temperature data are recorded. The peristaltic pump is then turned off, and the PCM inside the chamber is allowed to completely melt. Finally, the nuts are unscrewed on the two sides of the chamber, the PCM is poured out, and the 3-D finned tube is replaced with different parameters.

Fig. 2 shows the schematic and physical diagrams of the relevant size settings of the experimental chamber. The latter consists of a custom-made U-shaped pressure-resistant PC board and three rectangular pressure-resistant PC boards with a thickness of 15 mm. The rectangular boards are drilled with holes for the installation of fastening bolts, as shown in Fig. 2 (d). The U-shaped board is equipped with sealing gaskets on two sides to prevent any content leakage. The top rectangular pressure-resistant PC cover board is equipped with markings to facilitate

Table 1

Main equipment and parameters used in the experiment.

Equipment	Model	Manufacturer	Error
Data logger	34972A	Agilent	$\pm 0.004\%$
Peristaltic pump	LLS PLUS-S403	KAMOER	$\pm 0.001\text{ mL/min}$
Voltage stabilizer	TM-1	CHINT	$\pm 0.0001\text{ V}$
Transformer	TDGC2-1	CHINT	$\pm 0.5\text{ V}$
Power meter	PF9800	Yuanfang	$\pm 0.01\text{ W}$
Water tank	HH-W420	YITONG	$\pm 0.1\text{ }^{\circ}\text{C}$

the measurement and definition of the solid-liquid interface movement. During the experiment, the entire experimental chamber is wrapped with an asbestos blanket, having a thickness of 15 mm, to prevent the heat loss. The measurement equipment and parameters are shown in Table 2.

In order to conduct thermal management studies on lithium-ion batteries, their heating characteristics should be known. In this study, the CATL 40AH lithium-ion battery is used for testing. 8 thermocouples are evenly arranged on the surface of the cell: five on its positive surface, two at the ends of the electrode, and the remaining one at the bottom. The measurement point diagram is shown in Fig. 3.

Fig. 4 shows the position of the thermocouples placed on the tube surface. Nine thermocouples are strategically placed across the 3-D finned tube surface.

2.2. Structure parameter of the 3-D finned tube

A 3D diagram of the 3-D finned tube is shown in Fig. 5(a), and a photograph is shown in Fig. 5(b). It is deduced that the circumferential spacing (P_c) and varying the fin width (W) have similar effects on improving the heat transfer performance, and thus P_c is set as a constant in this study [29,30]. Table 3 presents the established structural characteristics for the 3-D finned tube.

2.3. Physical properties of PCM

The industrial paraffin, which is a widely used PCM, has adjustable phase change temperature, high thermal storage performance, and high chemical stability. In addition, it has a low cost and it can be easily obtained. Besides its low thermal conductivity, the battery and its pack shell do not face the problem of corrosion. Therefore, it is considered as the first choice for the thermal management phase change cooling materials for lithium-ion batteries [31].

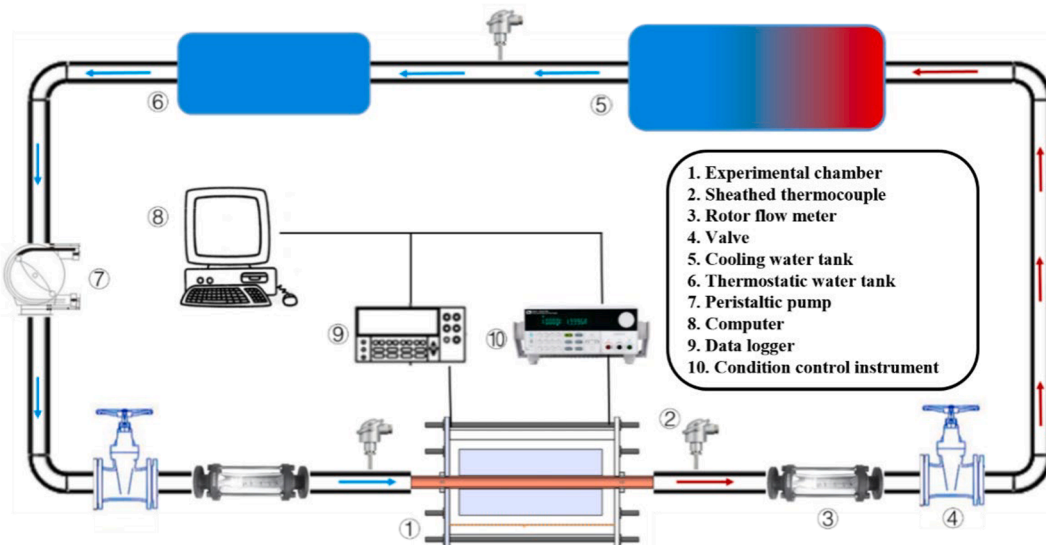


Fig. 1. Schematic diagram of the experimental setup.

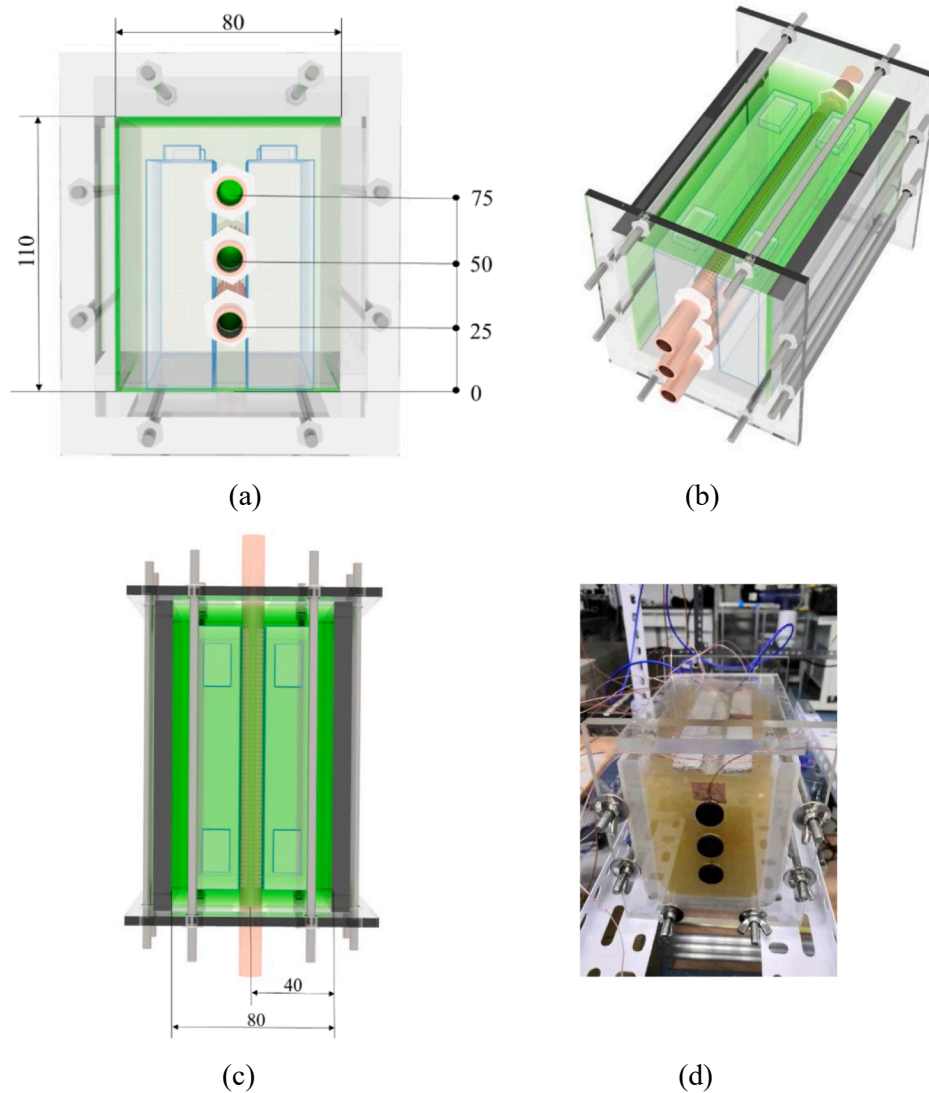


Fig. 2. Dimensions of the experimental chamber (mm):(a) front view, (b) oblique view, (c) top view, and (d) photograph.

Table 2

The measuring equipment and its main parameters.

Parameter	Acquisition equipment	Measuring range	Error
Wall temperature	Type T thermocouple	$-40 - 125\text{ }^{\circ}\text{C}$	$\pm 0.6\%$
Tube temperature	Type T thermocouple	$-40 - 125\text{ }^{\circ}\text{C}$	$\pm 0.6\%$
PCM temperature	Type T thermocouple	$-40 - 125\text{ }^{\circ}\text{C}$	$\pm 0.6\%$
Flow rate	Rotameter	$0 - 400\text{ mL/min}$	$\pm 1.5\%$
Water temperature	Type K armored thermocouple	$0 - 275\text{ }^{\circ}\text{C}$	$\pm 1.2\%$

In this study, a low-temperature phase change paraffin with 21 carbon atoms is purchased from Shanghai Aladdin Bio-Chem Technology Co., Ltd. The physical properties of the paraffin wax are summarized in Table 4. The phase change temperature for this paraffin wax is in the range of $39.5 - 41.5\text{ }^{\circ}\text{C}$. When the temperature of a specific region is in this range, the PCM within this region is in a state of solid–liquid coexistence.

2.4. Experiment on the battery and heating plate temperature increase

In this experiment, a heating plate is used instead of a CATL 40AH lithium-ion battery to stably control the heat source for a long time. The

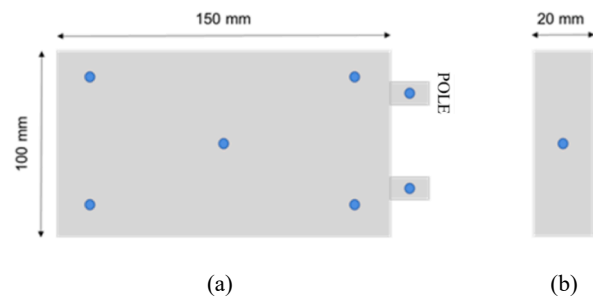


Fig. 3. Position of the thermocouples placed in the heating plate:(a) front view and (b) oblique view.

increases in the battery and heating plate temperatures under different working conditions are compared, and the feasibility of replacing the battery with the heating plate is verified.

The lithium-ion battery adopts the charging strategy of 1C constant current charging to 4.15 V to constant voltage charging. The cut-off current and voltage are 0.01C and 2.8 V, respectively. After charging, the single cell is placed for a certain period of time to cool it to room temperature, and the battery is then discharged at rates of 1, 1.5, 2, 2.5,

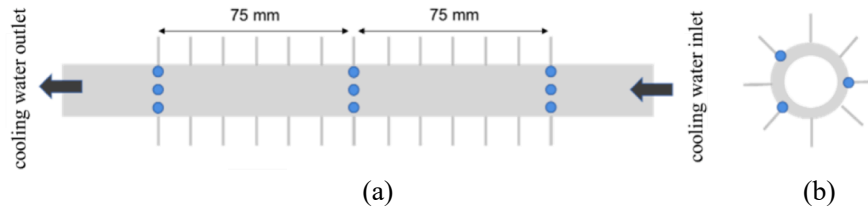


Fig. 4. Position of the thermocouples placed in the 3-D finned tube:(a) front view and (b) oblique view.

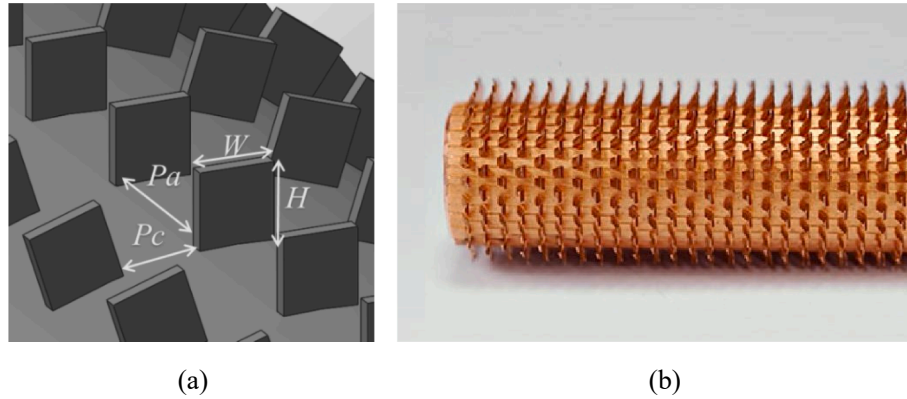


Fig. 5. 3-D finned tube: (a) 3D diagram and (b) photograph.

Table 3

Fin structure parameter settings of the 3-D finned tube.

Parameter	Data			
fin height H (mm)	1.0	2.0	3.0	4.0
fin width W (mm)	1.0	1.5	2.0	2.5
axial spacing P_a (mm)	2.0	3.0	4.0	5.0
circumferential spacing P_c (mm)	1.5			

Table 4

Thermophysical properties of PCM [32–34].

temperature ($^{\circ}\text{C}$)	ρ (kg/ m^3)	C_p (kJ/ (kg K))	latent heat (kJ/kg)	phase-change temperature ($^{\circ}\text{C}$)	λ (W/(m K))
39.5–41.5	780–789	2.1	220	39.5–41.5	0.251–0.256

and 3C. The battery is then cooled to room temperature for a certain period of time after each discharge. This process is repeated after charging using the same method. The temperatures on the surface of the battery cell during charging are measured and recorded using the Agilent 34972A data logger.

Fig. 6 shows a comparison between the temperature increase of the heating plate and that of the battery. The different discharge rates of 1.5, 2, 2.5, and 3C correspond to heating powers of 6, 12, 18, and 36 W, respectively. It can be seen from Fig. 5 that the curves of the temperature increase of the heating plate and the battery have the same trend. The maximum temperature increase gap is $\pm 4.14\%$, which indicates that the heating plate can replace the battery in the following thermal management study. In the thermal management performance test experiments, the heating plates, used instead of the battery, are placed in the experimental chamber of a transparent PC board processing. They are completely immersed in the low-temperature phase change paraffin wax, and the 3-D finned tubes are placed in the middle of two heating plates.

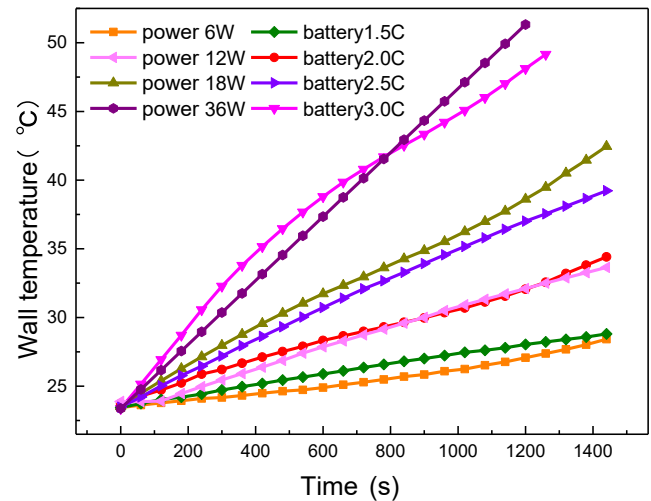


Fig. 6. Comparison between the temperature increase of the battery and the heating plate with different discharge rates.

2.5. Data processing

The PCM in the experimental chamber is divided into 4 parts along the direction perpendicular to the heating plate. More precisely, the solid–liquid interface is placed in different labels corresponding to different melting states. Based on the wall temperature and the melting state identified in the experimental chamber, the melting process is divided into 5 stages: solid state, liquid–solid ratio of 1:3, liquid–solid ratio of 2:2, liquid–solid ratio of 3:1, and close to liquid state. The close to liquid state indicates the case where the PCM in other places is completely melted and only a thin layer of it exists on the surface of the 3-D finned tube, resulting from cooling water flowing in the tube.

Since the study on the influence of the melting state on the thermal management performance presented in this paper tackles the steady state, it is necessary to first obtain different melting states of the PCM. At

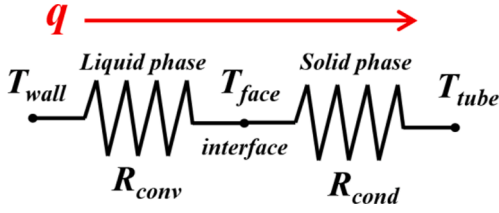


Fig. 7. Thermal circuit for heat conduction.

the beginning of the experiment, the heat power of heating plate was set to 36 W. Once the solid–liquid interface reached the designated position, the flow rate of the cooling water was initiated and upheld at 300 mL/min. The solid–liquid interface is adjusted by heating power of power meter and kept stable for 1 h. Meanwhile, the fluctuation of the measured value of the thermos couples is less than $\pm 0.5^\circ\text{C}$.

The main thermal resistance of heat transfer in PCM can be divided into two parts: the convective heat transfer thermal resistance in the liquid phase region and the thermal conductivity thermal resistance in the solid phase region. At the melting beginning, the paraffin wax is between solid state and liquid state, and the heat transfer process is mainly dominated by heat conduction. Therefore, in this state, liquid-state PCM and the solid-state PCM are considered together. The thermal resistance of the contact between the solid PCM and the metal wall surface is very small, and the thermal resistance of the metal heat exchanger in the experimental process is ignored.

The total heat transfer coefficient of the PCM combined with the 3-D finned tube is calculated as:

$$K = \frac{q}{A \cdot (T_{\text{wall}} - T_{\text{tube}})} \quad (1)$$

Where A present the tube surface area, and q is heat flux generate by heating plate, which can be directly read from the power meter. The T_{wall} and T_{tube} present the temperature of heating plate wall and 3-D finned tube. To streamline the analysis process, the difference between the temperature of the solid–liquid coexistence area and the average temperature of the heat exchanger base tube is considered as the heat transfer temperature difference in the solid phase region. The conduction thermal resistance of the solid phase region is calculated as:

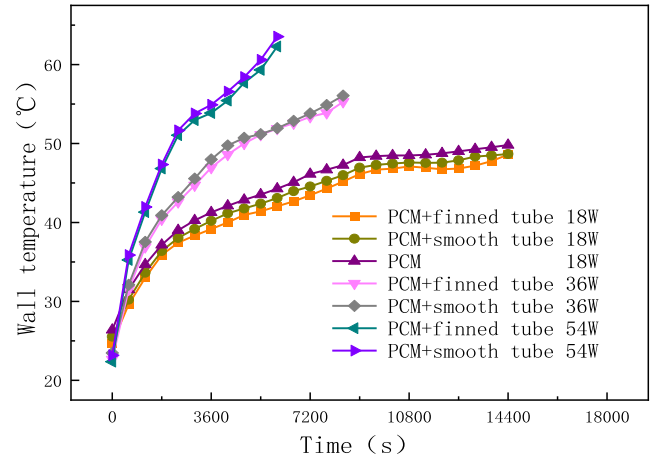
$$R_{\text{cond}} = \frac{(T_{\text{face}} - T_{\text{tube}})}{q} \quad (2)$$

where T_{face} present the temperature at the solid–liquid interface of the PCM. The thermal resistance of the convection is computed as:

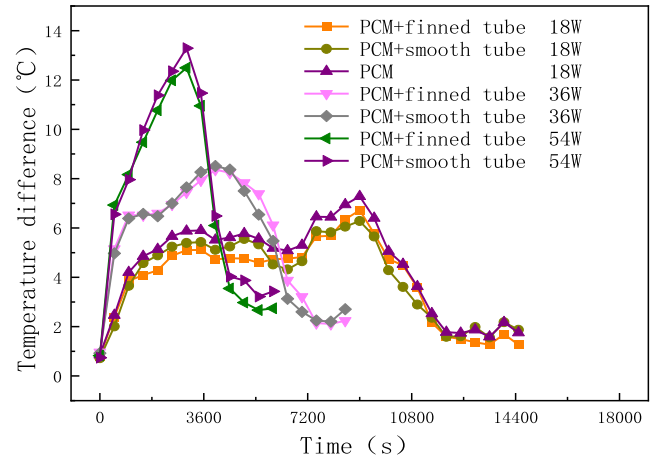
$$R_{\text{conv}} = \frac{(T_{\text{wall}} - T_{\text{tube}})}{q} - R_{\text{cond}} \quad (3)$$

The thermal circuit for Eqs. (1)–(3) is shown in Fig. 7.

In this study, the experiment is repeated three times with the same condition, and the average data is considered as the final result. The measured data are processed using the Moffat [35] uncertainty analysis method. The measurement uncertainties of K and R are calculated using Eqs. (4) and (5), as shown in Tables 1 and 2, respectively. The uncertainties of K and R are respectively $\pm 1.52\%$ and $\pm 1.51\%$, which indicates that the computed data are highly reliable for practical applications.



(a)



(b)

Fig. 8. Comparison between the thermal management effects of different combinations: wall temperature for different working conditions and (b) maximum temperature difference of the wall surface for different working conditions.

$$\Delta R = \left[\left(\frac{\partial R}{\partial q} \times \Delta q \right)^2 + \left(\frac{\partial R}{\partial T_{\text{wall}}} \times \Delta T_{\text{wall}} \right)^2 + \left(\frac{\partial R}{\partial T_{\text{tube}}} \times \Delta T_{\text{tube}} \right)^2 \right]^{0.5} \quad (5)$$

3. Results and discussion

3.1. Performance of different thermal management methods

Fig. 8 (a) shows the change of the maximum temperature of the heating plate surface with the increase of the heating time of the pure PCM, PCM combined with 3-D finned tube, and PCM combined with smooth tube for different heating powers and without the inflow of

$$\Delta K = \left[\left(\frac{\partial K}{\partial q} \times \Delta q \right)^2 + \left(\frac{\partial K}{\partial A} \times \Delta A \right)^2 + \left(\frac{\partial K}{\partial T_{\text{wall}}} \times \Delta T_{\text{wall}} \right)^2 + \left(\frac{\partial K}{\partial T_{\text{tube}}} \times \Delta T_{\text{tube}} \right)^2 \right]^{0.5} \quad (4)$$

cooling water. The reason why there is no cooling water in the tubes is to facilitate the comparison with the thermal management effect of pure PCM, because pure PCM can not pass cooling water. The fin height, fin width, axial spacing, and circumferential spacing of the 3-D finned tube are 1, 3, 2, and 1.5 mm, respectively. To ensure the long-term thermal management capabilities and conduct a thorough evaluation of performance of this system, the duration of the experiment exceeds the theoretical discharge rate duration corresponding to the applied heating power. This setup aims to explore the potential applicability in a variety of thermal management scenarios.

It can be seen that from Fig. 8 (a) the greater the heating power, the higher the maximum temperature increase rate on the surface of the heating plate. However, although the surface temperature of the heating plate of the PCM combined with 3-D finned tube and smooth tube continues to increase, the escalation pace of maximum temperature in the heating plate surface experiences a slow decrease during the melting process. The effect of the pure PCM thermal management is the worst. For all the heating powers, superior thermal management effects are observed when a 3-D finned tube is combined with PCM, compared with the smooth tube. Compared with the case where only PCM are used, the mean temperature of the wall surface when using the smooth tube is reduced by 1.81 °C. The average wall temperature using PCM combined with 3-D finned tube is further decreased by 1.02 °C compared with the case involving PCM combined with smooth tube.

Fig. 8 (b) shows the maximum temperature difference between the surface of the heating plate with the increase of the heating time without the inflow of cooling water. It can be seen that, when the PCM starts to melt, the maximum temperature difference on the surface of the heating plate first increases and then decreases. After stabilization, the maximum temperature difference between the heating plate wall surface of the experimental group employing the 3-D finned tube exhibits a temperature which is 0.56 °C (in average) cooler than the case employing smooth tube, and the maximum temperature difference is below 2.4 °C. It allows to maintain the temperature of the battery below 45 °C within 2 h when subjected to a heating power of 18 W. It also maintains its temperature uniformity at 2 °C for a long time.

In addition, when the 3-D finned tube and smooth tube are passed into the cooling water (24 °C and 360 mL/min), the 3-D finned tube demonstrated superior thermal management performance. For example, when the fin height, fin width, axial spacing, and circumferential spacing of 3-D finned tube are 4, 1, 1, and 1.5 mm respectively, the maximum temperature is reduced by 4.52 °C compared to the smooth tube, and with a maximum temperature difference of 2.32 °C. So according to the above analysis, it can be demonstrated that the 3-D finned

tube is more effective in managing heat source temperature equalization no matter with or without cooling water. It can be deduced from these obtained results that employing PCM combined with a 3-D finned tube enhances the thermal management ability.

3.2. Impact of the fin height on the thermal performance with different liquid-solid ratios

Fig. 9 shows the heat transfer coefficients obtained with different fin heights under different melting states. The width and axial spacing of the fins are both 1 mm, and the circumferential spacing is 1.5 mm. It can be seen that, when the fin height increases from 1 mm to 4 mm, the heat dissipation performance of the PCM combined with 3-D finned tube is enhanced below each melting state. The PCM heat transfer process is promoted with the increase of the fins height on the 3-D finned tubes. This is attributable to the enhanced effect on the natural convection of the liquid PCM. More precisely, the greater the fin height, the more pronounced the enhancement of the natural convection of the liquid PCM. In addition, increasing the height of the fins results in increasing the overall heat transfer area, which further boosts the heat transfer. Furthermore, when the liquid-solid ratio increases, the heat transfer coefficient first decreases and then increases. This is the result of the combined effect of the thermal conduction in the solid phase region and the convective heat transfer process in the liquid phase region.

Fig. 10 shows the thermal resistance of different liquid-solid ratios at different fin heights. It can be observed that, regardless of the fin height, when the liquid-solid ratio increases, the conduction thermal resistance of the solid PCM continues to decrease, and the convective thermal resistance of the liquid PCM continues to increase. This is due to the fact that, when the liquid-solid ratio increases, the thickness and volume of the liquid PCM both increase. During the phase transition, a natural convection exists in the liquid PCM. When the increase of the convective thermal resistance of the liquid PCM is greater than the decrease of the conduction thermal resistance, the total heat transfer coefficient decreases. In addition, when the increase of the convective thermal resistance of the liquid PCM is smaller than the decrease of the conduction thermal resistance, the total heat transfer coefficient increases. It is important to mention that, when the liquid-solid ratio is 3:1 and the fin height is 4 mm, the convective thermal resistance significantly increases (Fig. 10 (d)), and therefore the overall heat transfer coefficient significantly decreases (Fig. 9).

3.3. Impact of the fin width on the thermal management performance with different liquid-solid ratios

Fig. 11 illustrates the variation of the heat transfer coefficient for different fin widths under various liquid-solid ratios, with a fixed fin height of 3 mm. The axial spacing and height of the fins are both of 1 mm, and the circumferential spacing is 1.5 mm. Regardless of the liquid-solid ratio, when the fin width increases, the heat transfer efficacy of the system combining PCM with a 3-D finned tube first increases and then decreases. When the fin width increases, the cross-sectional area of the fins also increases. In addition, the heat transfer capacity of the fins is promoted by the expansion of the thermal exchange surface. Furthermore, the increase of the fin width results in increasing the interface between the fins and the PCM, which significantly increases the heat transfer efficiency during the preliminary phase. However, when the fin width reaches a certain extent, the number of gaps along the circumferential fins is reduced. This impedes the flow of the liquid phase in the PCM between the fins, and increases the distance of heat transfer from the central part of the PCM to the fins. Hence, when the fin width continues to increase, the heat transfer efficiency starts to decrease in each melting state of PCM.

Fig. 12 shows the thermal resistance of the PCM combined with 3-D finned tubes with different fin widths at different melting states. It can be clearly seen that under each fin width parameter, when the

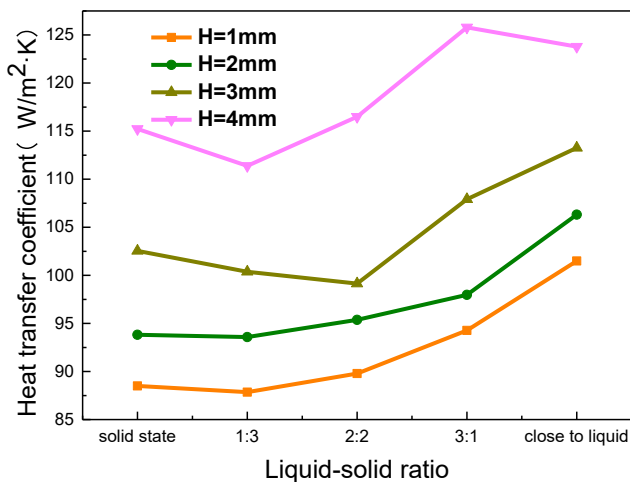
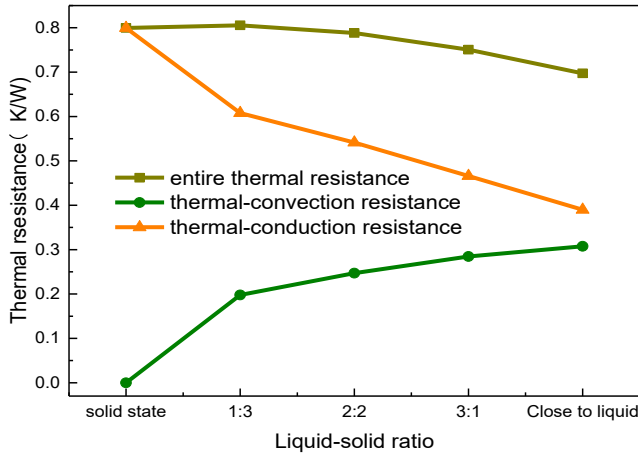
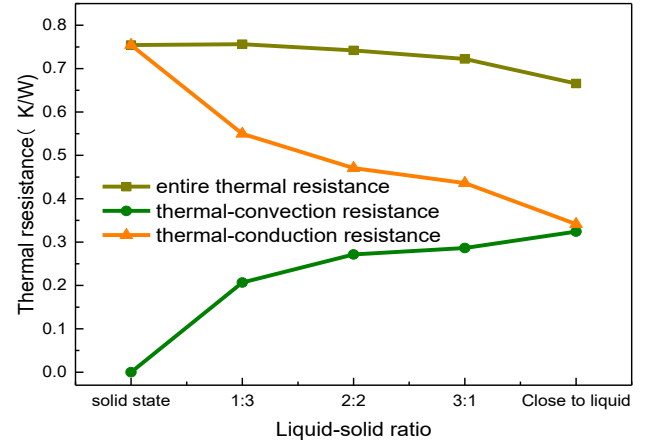


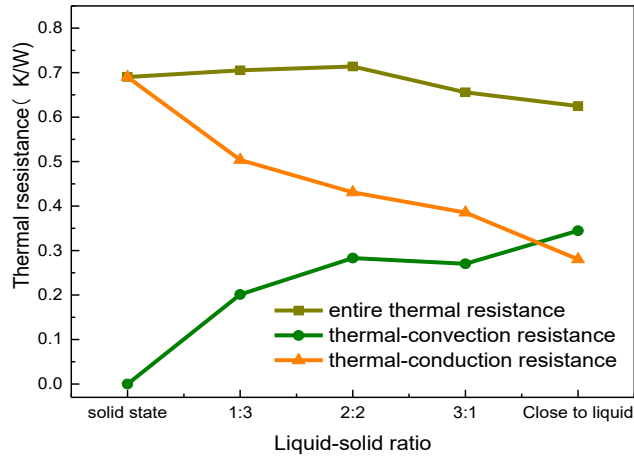
Fig. 9. Heat transfer coefficient for different fin heights and liquid-solid ratios, with fin width and axial spacing of 1 mm, and circumferential spacing of 1.5 mm.



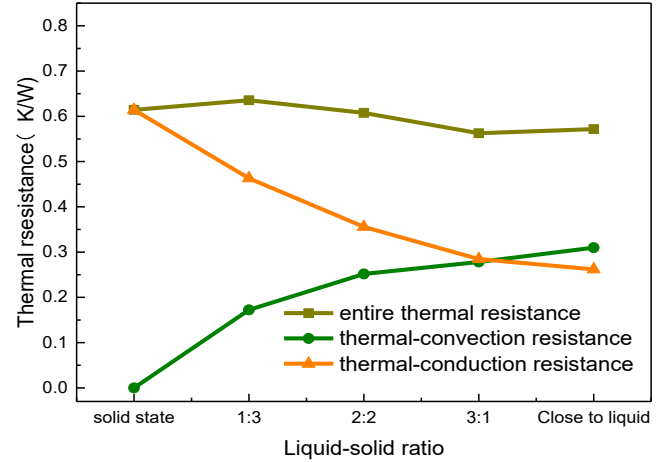
(a) fin height of 1 mm



(b) fin height of 2 mm



(c) fin height of 3 mm



(d) fin height of 4 mm

Fig. 10. Thermal resistance of different liquid-solid ratios at different fin heights.

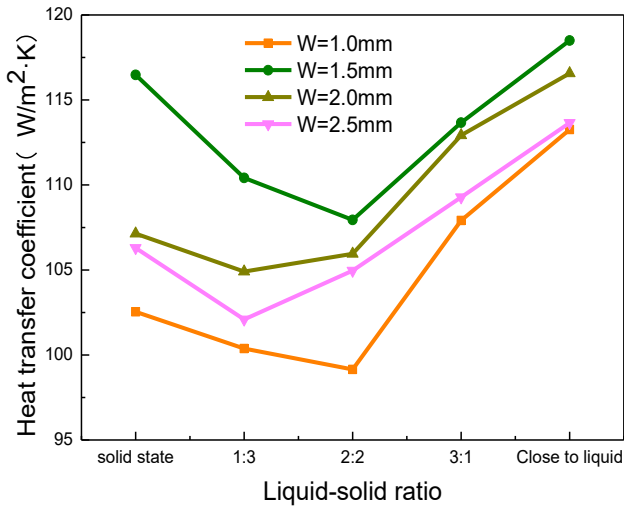
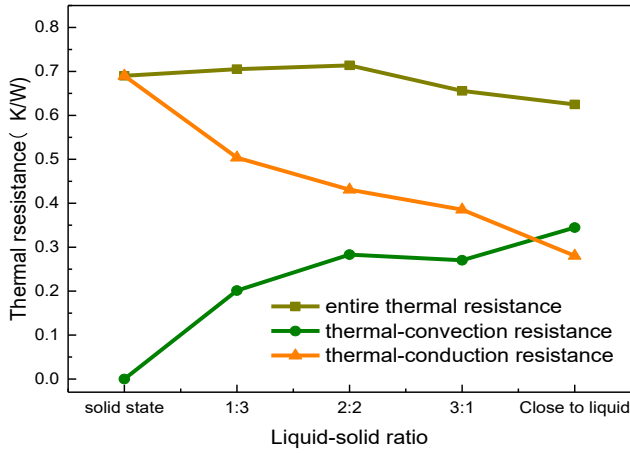


Fig. 11. Heat transfer coefficient for different fin widths and liquid-solid ratios, with a fin height and an axial spacing of 1 mm, and a circumferential spacing of 1.5 mm.

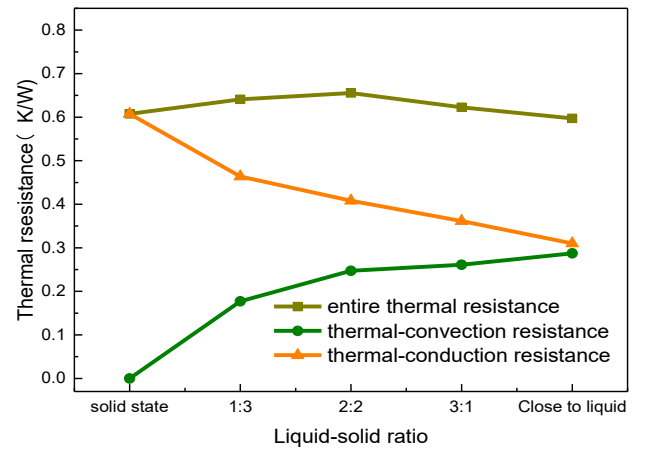
liquid-solid ratio increases, the thermal resistance of thermal conduction continues to decrease, while the thermal resistance of convection continues to increase. In the initial melting stage, the convective thermal resistance of PCM always increases by a larger amount compared with that of the conduction thermal resistance decrease. Therefore, the overall heat transfer performance decreases. At this stage, the increase of the convective thermal resistance is gradually attenuated. As a result, the overall heat transfer performance starts to increase. Therefore, when the liquid-solid ratio increases, the overall heat transfer performance first decreases and then increases.

3.4. Impact of the axial spacing on the thermal management performance with different liquid-solid ratios

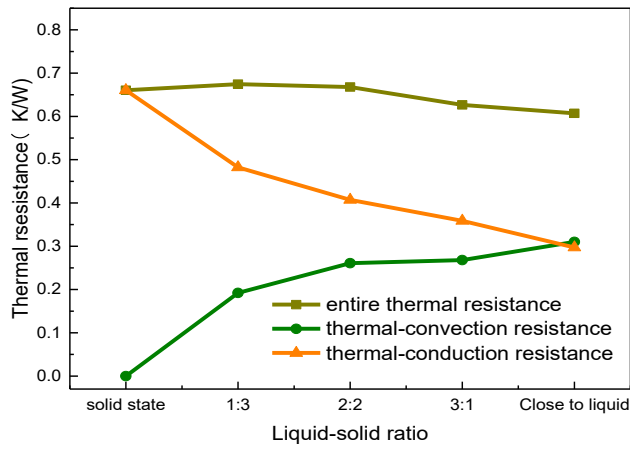
Fig. 13 shows the heat transfer coefficient of the PCM combined with 3-D finned tube featuring distinct axial spacings for a different liquid-solid ratios. The height and width of the fins are both of 1 mm, and the circumferential spacing is 1.5 mm. It can be seen that, when the axial spacing increases, the heat transfer coefficient first increases and then decreases in any liquid-solid ratio. After reaching 4 mm, when the axial spacing further increases, the heat exchange coefficient significantly decreases. In the liquid PCM, heat transfer occurs predominantly through natural convection. The increase of the axial spacing at small



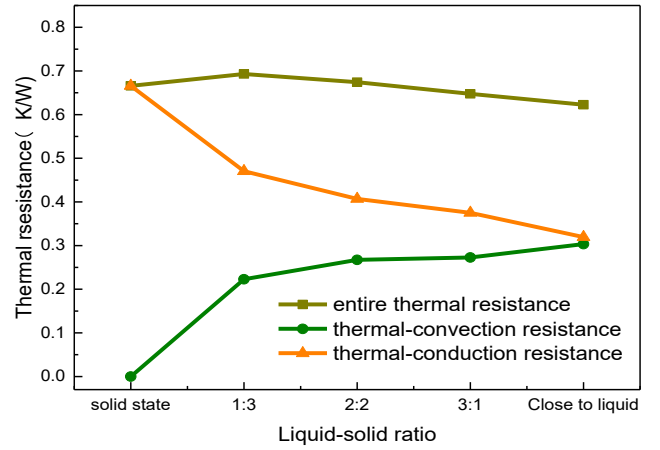
(a) fin width of 1 mm



(b) fin width of 1.5 mm



(c) fin width of 2 mm



(d) fin width of 2.5 mm

Fig. 12. Thermal resistance for different liquid solid ratios at different fin widths.

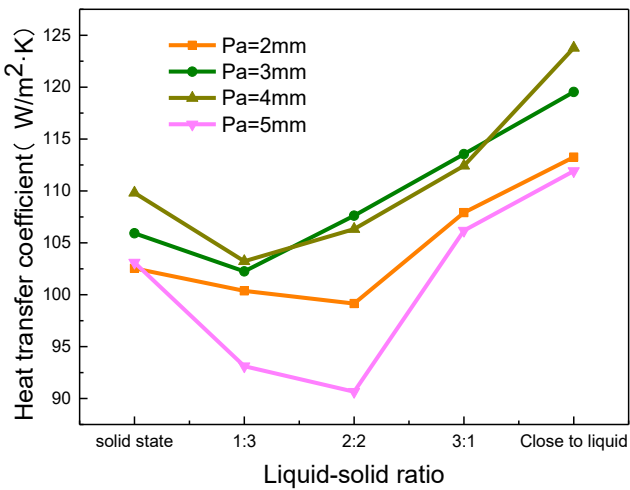
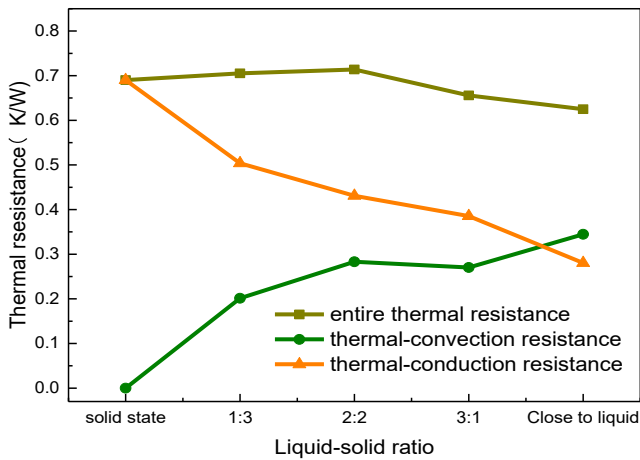


Fig. 13. Heat transfer coefficient for different axial spacings and liquid-solid ratios, with fin height and width of 1 mm, and circumferential spacing of 1.5 mm.

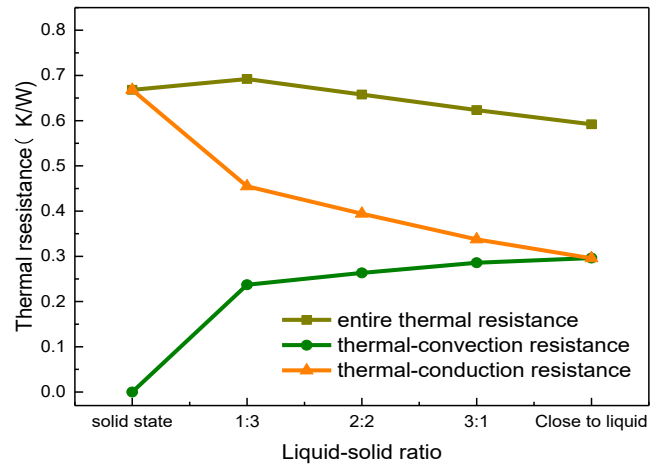
values may improve the heat transfer performance by enlarging the natural convective action space of the liquid PCM. On the contrary, if the axial spacing significantly increases, the increase of the heat transfer path from the PCM to the fins and the reduction of the total surface area of the fins may start to exert a dominant effect, which results in a decreasing the heat transfer performance. Hence, when the fin axial spacing increases, the total heat transfer efficiency first increases and then decreases in each solid-liquid ratio.

When the liquid-solid ratio increases, the overall trend of the heat transfer coefficient remains first decreasing and then increasing. Fig. 14 shows the thermal resistance of the PCM combined with 3-D finned tube with different liquid-solid ratios and different axial spacings. At the beginning of the melting process, the conduction thermal resistance decreases while the convective thermal resistance significantly increases. At this point, the overall heat transfer coefficient decreases. When the liquid-solid ratio increases, the 3-D finned tubes affect the liquid PCM, the convection thermal resistance increases by a larger amount compared with the decrease of the conduction thermal resistance, and the thermal resistance starts to decrease.

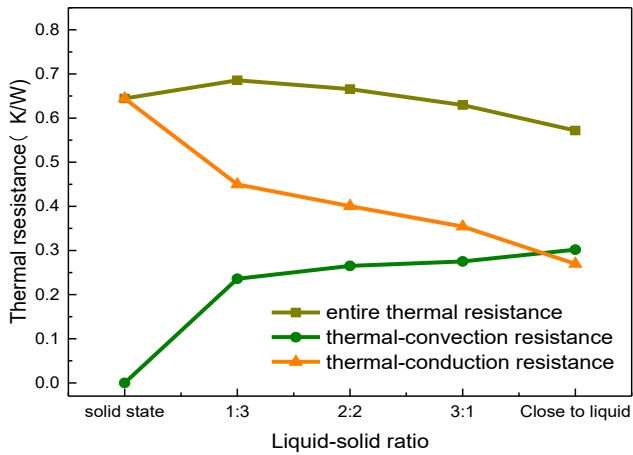
In summary, the thermal transfer capacity of the PCM combined with the 3-D finned tube is subject to the convective heat transfer within the liquid phase as well as the heat conduction within the solid phase. The optimal thermal management range of PCM can be determined by different structural parameters of the 3-D finned tube and PCM melt states. It can be deduced from the obtained results that regardless of the



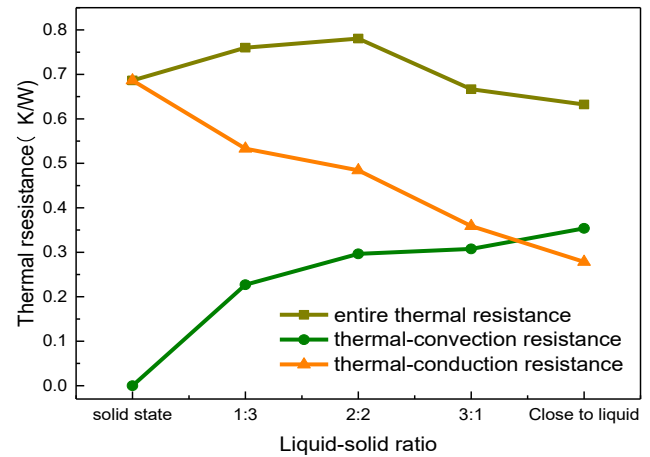
(a) axial spacing of 2 mm



(b) axial spacing of 3 mm



(c) axial spacing of 4 mm



(d) axial spacing of 5 mm

Fig. 14. Thermal resistance for different liquid solid ratios and different axial spacing.

change of the fins structural parameters, the heat dissipation effects are better when the liquid–solid ratio is greater than 2:2. Therefore, within this range, the heat absorption and dissipation characteristics of the PCM can be most effectively used to maximize the cooling performance of the cell.

4. Conclusion

In this paper, a novel thermal management method, combining the 3-D finned tube with the PCM, is proposed. The thermal transfer efficiency of the PCM combined with 3-D finned tube heat exchangers is explored across different states of melting and fin structure parameters. The obtained results show that the solid–liquid ratio is an important factor affecting the heat transfer process. In addition, the heat transfer performance is positively associated with the increase of the fin height of the 3-D finned tube in each solid–liquid ratio of PCM. More precisely, it increases with the increase of the fin height. When the dimensions of either the fin width or the axial spacing in the 3-D finned tube increase, the thermal transfer efficiency first increases and then decreases in each melting state of PCM. Moreover, the combination of the PCM and the 3-D finned tube enhances the thermal management performance when the liquid–solid ratio is greater than 2:2. Thus, to optimize the thermal management performance of a lithium-ion battery, a 3-D finned tube heat exchanger with large fin height should be used. This allows to maintain the liquid–solid ratio of the PCM above the threshold of 2:2.

The results obtained in this study are of great significance for the design of BTMS and the application of PCM in thermal management.

CRediT authorship contribution statement

Zhao-ran Li: Writing – review & editing, Writing – original draft, Data curation. **Gao-ning Liang:** Methodology, Investigation, Data curation, Conceptualization. **Yu-dong Ding:** Software, Project administration. **Qiang Liao:** Resources. **Xun Zhu:** Resources. **Min Cheng:** Writing – review & editing, Visualization, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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